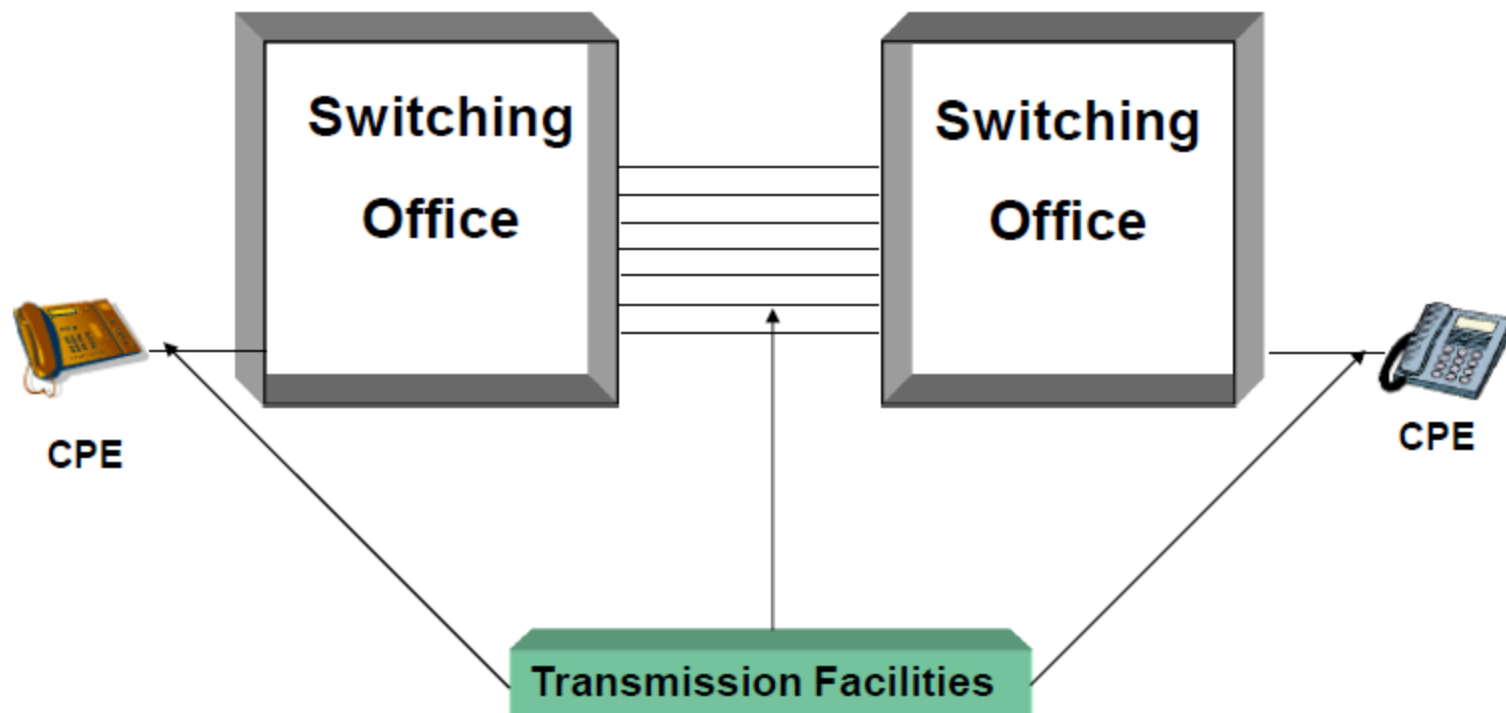


Public Switched Telephone Network (PSTN)

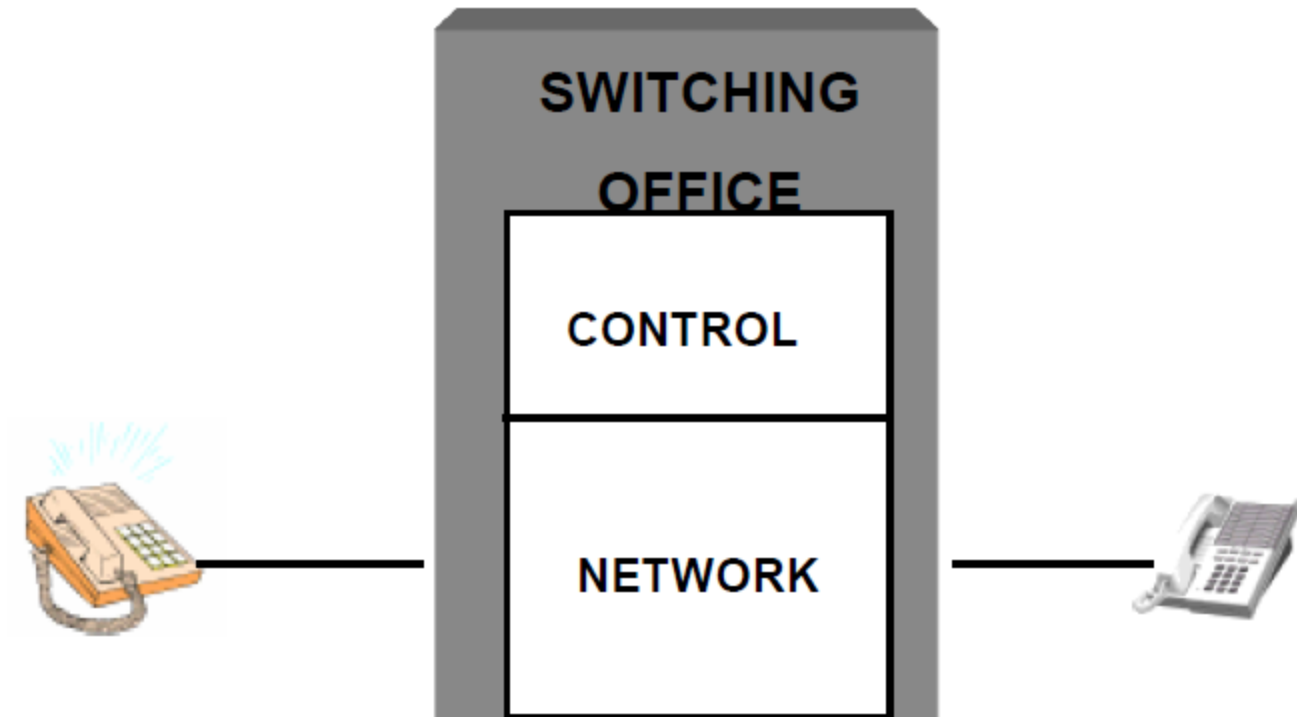


Public Switched Telephone Network (PSTN)

Major Components of the Public Switched Telephone Network (PSTN):

- **Switching Offices**
 - **Transmission facilities**
 - **Customer Premise Equipment (CPE)**
-

Switching Offices



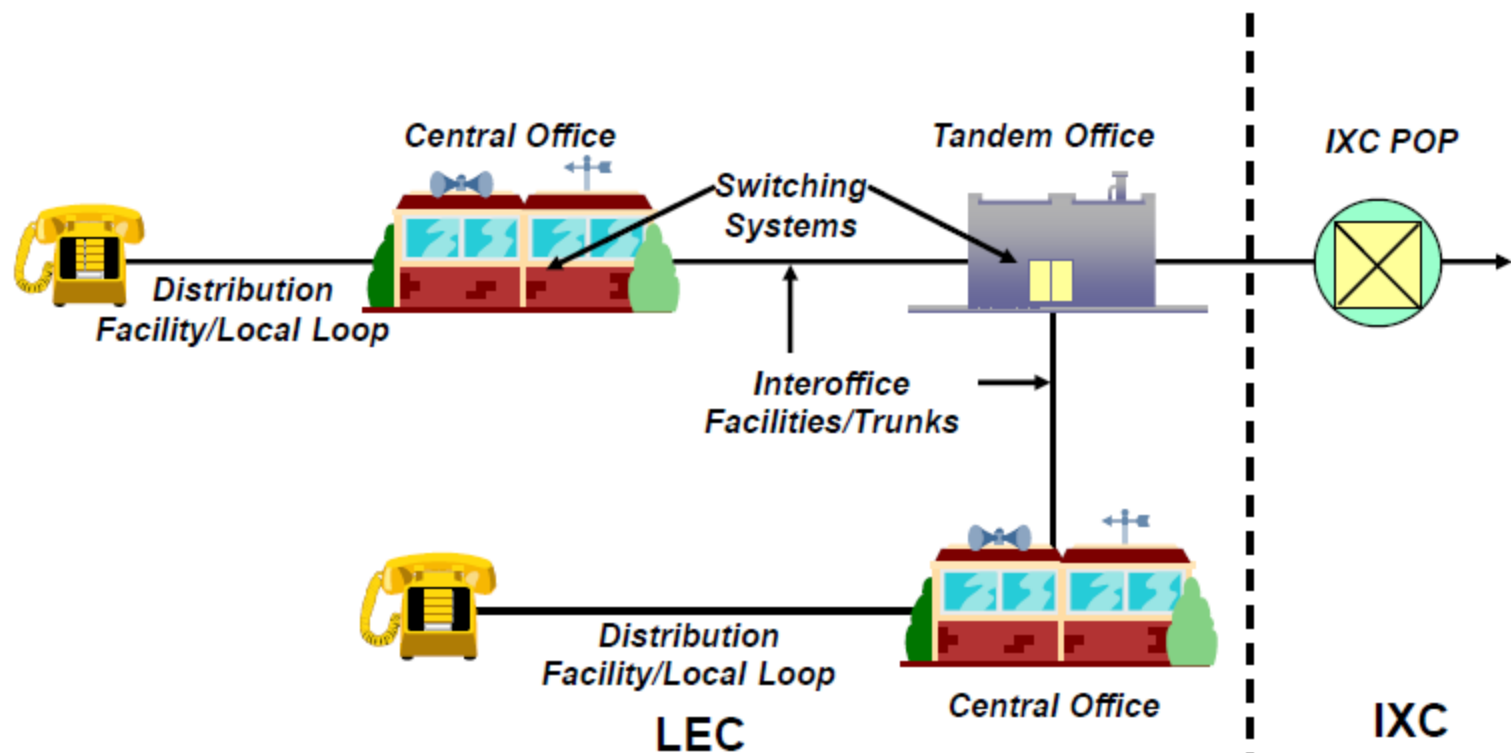
Public Switched Telephone Network (PSTN)

- Each local exchange consists of a central office (CO) which provides PSTN connection to the Customer Premises Equipment (CPE) which may be an individual phone at a residence or a private branch at a place of business.
 - The CO may handle as many as million telephone connections.
 - The CO is connected to a tandem switch which in turn connects the local exchange to the PSTN.
 - The tandem switch physically connects the local telephone network to the point of presence of trunked radio lines provided by one or more IXCs.
-

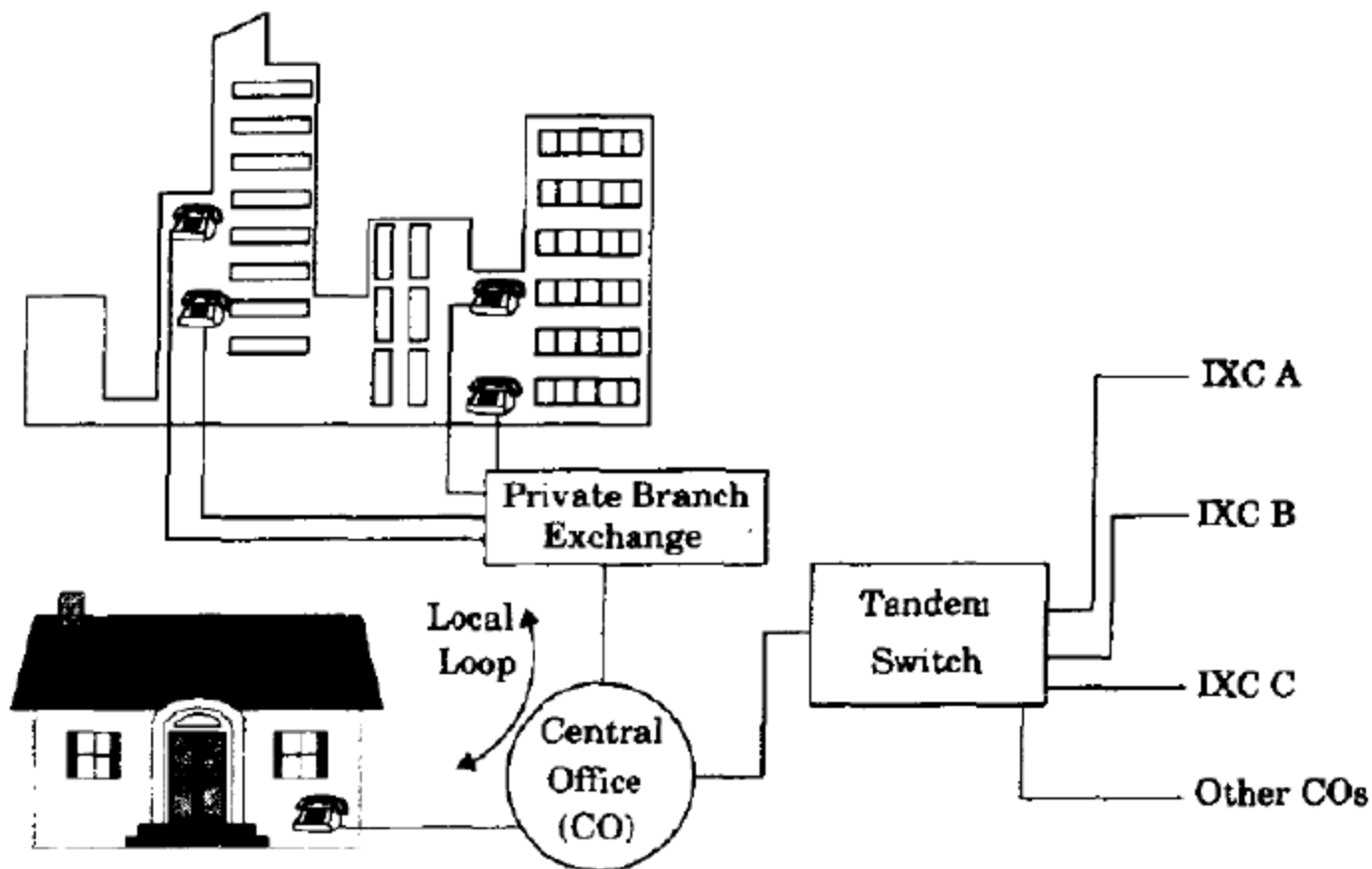
PSTN (Public Switched Telephone Network)

- In PSTN, each city or geographic grouping of towns is called a Local Access and Transport Area (LATA) .
 - Surrounding LATAs are connected by a company called a Local Exchange Carrier (LEC).
 - A long distance telephone company collects toll fees to provide connections between different LATAs over its long distance network. These companies are referred as interexchange carriers.(IXCs)
 - IXCs own and operate large fiber optic and microwave radio networks which are connected to LECs throughout a country.
-

Typical LEC Network

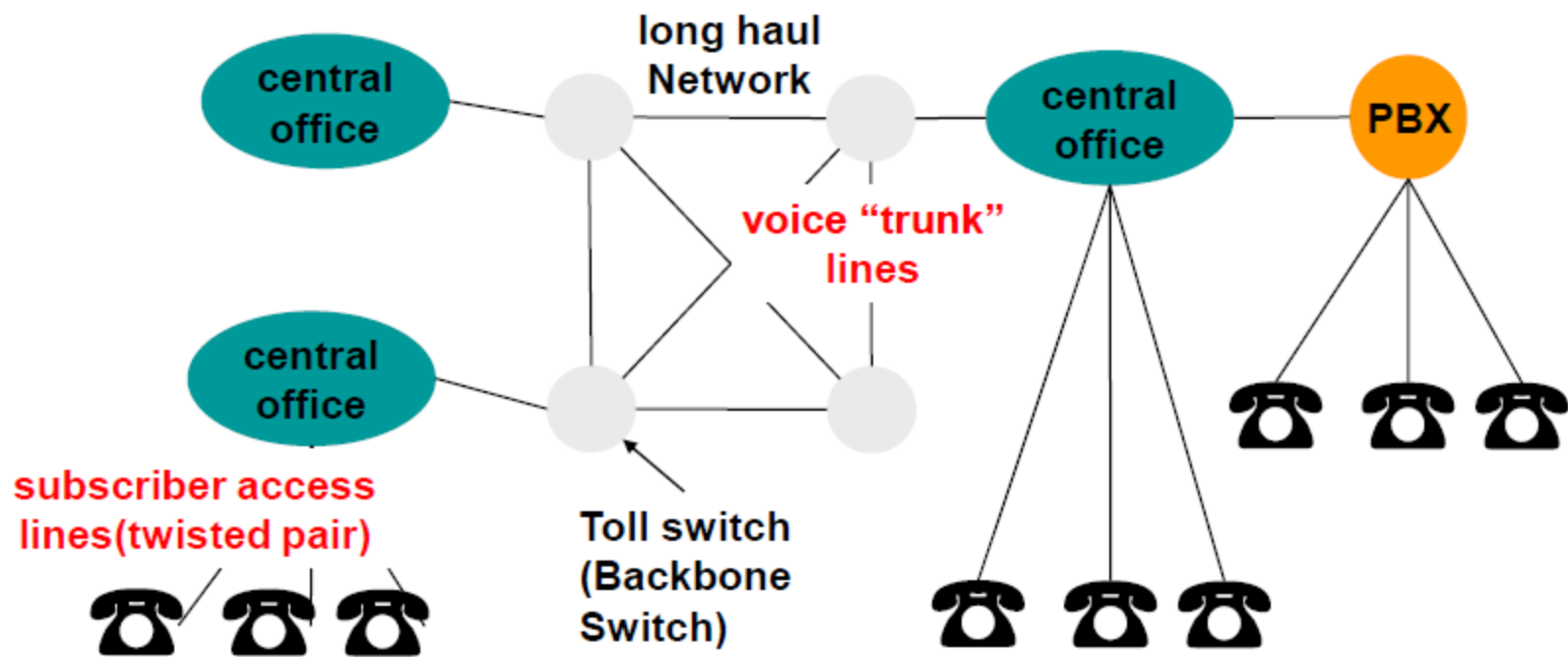


Local Landline Telephone Network

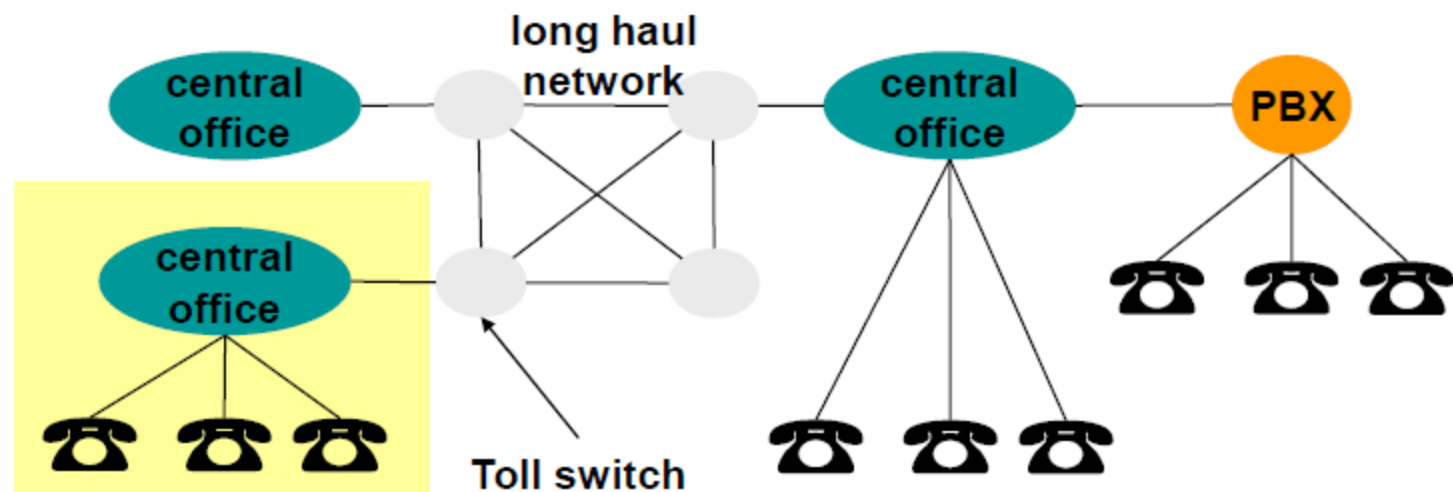


Telephone Network

- r created 1876
- r currently a global Infrastructure



Central Office and Local Loop



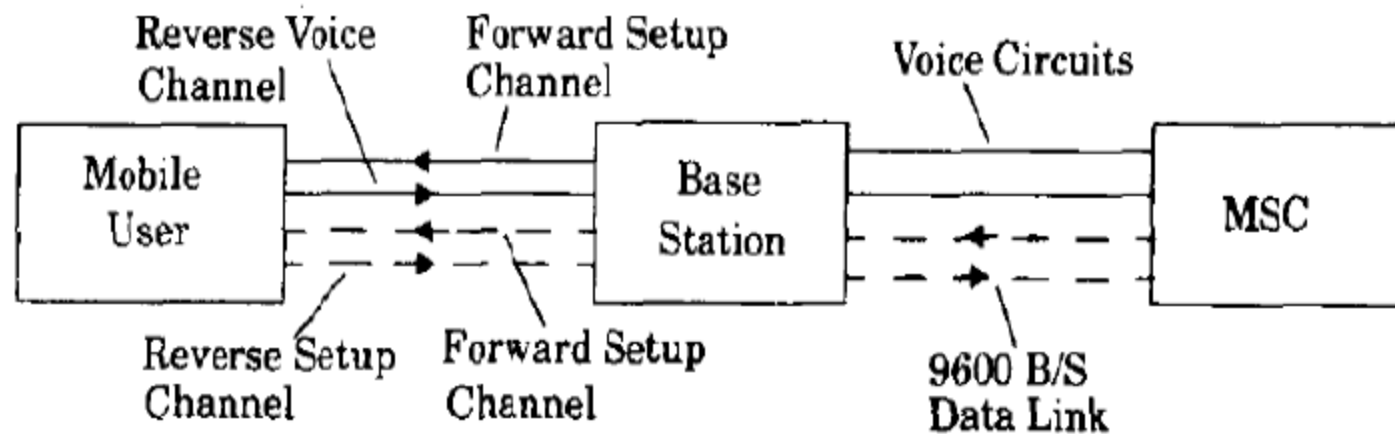
- Each phone user (**subscriber**) has direct connection to switch in central office (**local loop**)
- local loop has length 1 - 10 km
- switches in central office called (**local**) **exchanges**
- company providing local telephone service called **local exchange carrier** or **LEC**.

Development of Wireless Networks

FIRST GENERATION NETWORKS

- An example is AMPS (Analog Mobile Phone Service)
 - Used Frequency Modulation.
 - MSC has complete system control like
 - ✓ Maintaining Mobile Related Information.
 - ✓ Performing all network management functions like call handling ,processing, billing, fraud detection within the market
-

COMMUNICATION SIGNALLING BETWEEN MOBILE ,BS AND MSC IN 1st GENERATION



2nd GENERATION

- 2 G uses digital modulation ,advanced call processing capabilities and new network architectures reducing computational burden on MSC.
 - Examples are-GSM,TDMA&CDMA US Digital Standards.
 - GSM introduced the concept of BSC (Base station Controller) thereby standardizing data interface between BSC and MSC
-

2nd GENERATION

- 2 G systems use digital voice coding & digital modulation techniques.
 - Mobile Station assumed greater control functions like
 - ✓ MAHO (Mobile Assisted Hand off)
 - ✓ Received Power Reporting
 - ✓ Adjacent Base Station Scanning
 - ✓ Data Encoding
 - ✓ Encryption
-

3rd GENERATION

- Aim of 3 G is to provide a single set of standards that can meet a wide range of wireless applications and provide universal access throughout the world.
 - 3 G uses Broadband Integrated Services Digital Network (B-ISDN)
 - 3 G networks carry many types of information (voice, data & video) operating in varied regions
-

Paradigm From 1G to Beyond 3G

